Contents

Reader-facing sound-change pilot

This assembled pilot PDF is for checking the first reader-facing rewrite layer of the sound-change half.

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3. Nasal dissimilation

Velar palatalization before front vowels

1. Historical discussion

Luick places the change inside a broad early palatalizing movement. Under the heading “Frühe Verschiebungen in palataler Richtung,” he treats English k and g before bright vowels together with the larger field of palatal effects (Luick 1914--1940, 157, §168).

Campbell narrows the picture by distinguishing plain velars from the especially palatal-prone sk cluster. His remark that "[sk] is more prone to palatalization and assibilation than [k]" is brief, but it makes clear that different members of the larger palatal field need not behave identically (Campbell 1959, 278, §440).

Ringe and Taylor make the chronological relation still clearer. When they write that "after initial velars and *sk had been palatalized" West-Saxon diphthongization follows, plain velar palatalization becomes an earlier consonantal stage presupposed by later vowel developments (Ringe and Taylor 2014, 215, §6.5.1).

2. Development of the discussion

Taken together, these accounts show a gradual tightening of focus. Luick treats palatalization as a broad early movement. Campbell distinguishes more sharply between plain velars and the sk complex. Ringe and Taylor then place the plain velar change in an explicit sequence that leads forward to later West-Saxon diphthongization. The literature therefore supports two claims at once: the change belongs to a larger palatalizing environment, but it also needs to be kept distinct from neighboring processes if the sequence of developments is to be described accurately.

3. Formalization in the present project

The implementation formalizes the change with one helper definition for plain k and a second block for plain g:

```
define OEVelarPalatalizationKFront [
    {*k} -> {*ʧ} || .#. _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || _ [{*i} | {*ī}],
    {*k} -> {*ʧ} || _ {*ī́},
    {*k} -> {*ʧ} || [{*i} | {*ī}] _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || {*ī́} _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || [{*i} | {*ī}] _ .#.,
    {*k} -> {*ʧ} || {*ī́} _ .#.
] .o. [
    {*k} {*k} -> {*ʧ} {*ʧ} || _ {*j}
] .o. [
    {*k} -> {*ʧ} || _ {*j}
] ;

define OEVelarPalatalization [
    OEVelarPalatalizationKFront
] .o. [
    {*g} -> {*ʤ} || _ EnglishStarFrontVowel,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ .#.,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ EnglishStarFrontVowel,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ [EnglishStarConsonant - {*j}],
    {*g} {*g} -> {*ʤ} {*ʤ} || _ {*j}
] .o. [
    {*g} -> {*ʤ} || _ {*j}
];
```

In prose, the rule does exactly what the title suggests: it palatalizes plain k and g in front-vocalic and j-adjacent environments. The point of writing it as a separate rule is not to deny the larger palatal field, but to make the relative order of plain-velar palatalization, sk-palatalization, and umlautal developments testable.

4. Chronological placement

The chronology can be tested by moving the rule in either direction.

Placed too early, before the syncope that prepares the consonant cluster, it breaks the derivation of *stretch*. With PGmc **strákkijaną* in the wrong order, the model produces *strecċan* instead of the expected Old English *streċċan*.

Placed too late, after i-umlaut, it over-palatalizes forms such as *cow* and *lung*. PGmc **kūi* then yields *ċȳ* instead of expected *cȳ*, and PGmc **lúnganjō* yields *lunġen* instead of expected *lungen*.

That is the reason for the rule's present position: it must come after the syncope that prepares forms like *stretch*, but before the umlautal stage that would otherwise create the wrong palatalized outputs in *cow* and *lung*.

5. Consequences for reconstructed forms

Once the rule is in place, plain velars before front vowels and `j` no longer remain plain. They become the palatal outcomes that later chapters presuppose. That matters not only for dictionary-like forms such as *cild* or *dœg*, but also for the broader relation between consonantal palatalization and later vowel-fronting processes (Luick 1914--1940, 157, §168; Campbell 1959, 278, §440; Ringe and Taylor 2014, 203–15, §§6.4.1, 6.5.1).

In other words, the rule is consequential because it creates the consonantal environment inherited by the umlaut chapter. Without it, later reconstructed forms are not merely shifted in date; they come out with the wrong consonant quality.

6. Remaining cautions

This change belongs to a wider palatalizing environment, but the evidence does not require that every neighboring palatal process be merged with it. `sk` belongs to a related but not identical development, and the later umlautal material poses a different historical problem. The left-hand relation to Sievers-law syncope is likewise specific rather than expansive: the *stretch* evidence shows a real dependency without turning the feeder process into a coequal sound law of the same scope.

The Old English i-umlaut and its West Saxon right edge

1. Historical discussion

Luick gives the change its traditional scale:

> Der wichtigste Fall von palataler Beeinflussung … war die Veränderung der urenglischen Vokale durch i oder j der Folgesilbe.
>
> In context, this means that i-umlaut is treated as the central case of palatal influence in early English.

(Luick 1914--1940, 166–67, §182)

Campbell gives the most compact classical formulation in English when he writes that "the process known as i-umlaut or i-mutation operates on practically all the sounds which it could theoretically affect in OE" (Campbell 1959, 69, §190). Hogg continues in the same vein: "we come now to a change which is almost as uncontroversial as it is important" (Hogg 1992, 112). Taken together, these statements leave little doubt that i-umlaut is one of the central Old English vowel changes.

The narrower palatal-diphthongal material is described differently. Ringe and Taylor treat West-Saxon diphthongization after initial palatals as a distinct process (Ringe and Taylor 2014, 215, §6.5.1), and Fulk is even more explicit about its chronological delicacy when he calls it "diphthongization by initial palatal consonants (which precedes front umlaut but not breaking)" (Fulk 2018, 74, §4.13).

2. Development of the discussion

The sequence of discussion is fairly clear. Luick, Campbell, and Hogg all give i-umlaut primary importance. Ringe and Taylor and Fulk then help separate that major change from the narrower West-Saxon diphthongization that stands beside it. The literature therefore establishes a large, system-wide umlautal change and a narrower adjoining process affecting words after initial palatals. That distinction matters because the two processes act in different environments and produce different lexical consequences.

3. Formalization in the present project

The implementation keeps the broad umlaut rule and the narrower West-Saxon diphthongization rule separate:

```
define OEIUmlautFronting [
    {*a} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ā} -> {*ǣ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*e} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*o} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ō} -> {*ē} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*u} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ū} -> {*ȳ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*á} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*é} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ó} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ú} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];

define OEIUmlautRaising [
    {*æ} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];

define OEIUmlautDiphthong [
    {*ea} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
```

```
    {*ēa} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*io} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*īo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*eo} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éa} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éo} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ío} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];

define OEIUmlaut OEIUmlautFronting
    .o. OEIUmlautRaising
    .o. OEIUmlautDiphthong;

define OEWsPalatalDiphthongization [
    {*æ} -> {*ea} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant | EnglishPa
    {*ǣ} -> {*ēa} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant | EnglishPa
    {*e} -> {*ie} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant | EnglishPa
    {*ē} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant | EnglishPa
    {*é} -> {*íe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant | EnglishPa
    {*ḗ} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant | EnglishPa
];
```

In prose, the distinction is straightforward. `OEIUmlaut` performs the broad fronting, raising, and diphthongal adjustments caused by following `i/j`. `OEWsPalatalDiphthongization` is much narrower: it applies at the left edge of a word after an already palatal consonant and creates specifically West-Saxon diphthongal outputs.

4. Chronological placement

The present chronology is fixed by concrete lexical failures in both directions.

If i-umlaut is moved too early, before velar palatalization, the outputs of *cow* and *lung* become over-palatalized. PGmc **kūi* yields *ċȳ* instead of expected *cȳ*, and PGmc **lúnganjō* yields *lunġen* instead of expected *lungen*.

If i-umlaut is moved too late, or if West-Saxon diphthongization is moved too early, the local right edge breaks around *gift* and *sheath*. PGmc **géftiz* then yields *ġieft* instead of expected *ġift*, and PGmc **skáiθiz* yields *sċǣþ* instead of expected *sċēaþ*.

The crucial limit is on the later side of the West-Saxon diphthongization rule. No tested lexical item fixes a narrower later boundary. There is therefore no warrant for turning that negative result into a historical claim that the rule must stand before the final stages of the sequence.

5. Consequences for reconstructed forms

The chapter changes the shape of reconstructed forms on a large scale. In the umlaut rule proper, back vowels front and high vowels raise under following `i/j`; this is why forms such as *cȳ* and *lungen* belong to the same chapter as more familiar textbook examples like *giest* and *giefan* (Campbell 1959, 69–72, §§190–191; Fulk 2018, 61–63, §4.7).

The West-Saxon follower has a narrower consequence. It produces the diphthongal surface forms expected in words such as *giefan* and *sceap*, but it does so only after the broader umlautal setting is already in place (Ringe and Taylor 2014, 215, §6.5.1; Hogg 1992, 108). That is why the implementation treats it as a narrower follow-on process rather than as a second main change of equal scale.

6. Remaining cautions

The West-Saxon diphthongal material is genuine, but it is narrower, more dialect-specific, and less stably placed in the broader chronology than the umlautal change itself. The evidence is therefore strongest when it is used to distinguish the two processes rather than to collapse them into one undifferentiated chapter. The available tests fix the left edge well, but they do not license a sweeping claim about the far right of the whole Old English sequence.

Nasal dissimilation

1. Historical discussion

Luick preserves individual outcomes such as “enitre ‘einjährig (aus *anwintri)” without building a separate chapter around them (Luick 1914--1940, 166). Campbell likewise reaches forms such as *heofon* in a discussion of suffixal variation rather than in any special section on nasal dissimilation (Campbell 1959, 155). Hogg mentions *heofon* in the course of his account of back mutation, again without isolating a separate law (Hogg 1992, 112).

Fulk supplies the clearest general formulation: “In the cluster mn, the first consonant tends to lose its nasality by dissimilation, though the results are hardly regular” (Fulk 2018, 121, §6.11). Ringe and Taylor stay close to the lexical evidence and note that *enetre* reflects “loss of the second *n by dissimilation” (Ringe and Taylor 2014, 282).

2. Development of the discussion

The discussion therefore develops from scattered lexical observations to a more explicit but still cautious generalization. Luick preserves the kind of form the rule is meant to capture. Campbell and Hogg show that related outcomes enter the handbooks, but only incidentally, as part of larger accounts of other changes. Fulk makes the recurrent `mn` tendency explicit, while Ringe and Taylor provide an exact lexical case in *enetre*. What emerges is a limited but recurring dissimilatory pattern rather than a sound change of the same scope as the major Old English vowel laws.

3. Formalization in the present project

The implementation isolates the change very narrowly:

```
define OENasalDissimilation [
    {*m} -> {*f} || EnglishStarShortVowel _ EnglishStarShortVowel {*n} [EnglishStarShort
];
```

The code targets medial `m` in a short-vowel environment before a following syllable containing `n`, and it rewrites that `m` as `f`. This is a stricter and more explicit formulation than the handbooks usually give. That explicitness is useful for the implementation, but it should not be mistaken for evidence that the traditional scholarship isolates exactly the same rule in exactly the same shape.

4. Chronological placement

The current chronology evidence is negative in both directions.

When the rule is moved earlier, the present tests find no lexical breakpoint before the inherited West-Germanic material that lies to the left of the Old English sequence. When it is moved later, they likewise fail to identify a narrower lexical boundary before the far right edge of the tested Old English sequence.

No comparable pair of lexical failures fixes a narrower slot here. The present tests do not identify sharper evidence for an earlier or later position within the Old English sequence.

5. Consequences for reconstructed forms

Even so, the rule has real interpretative consequences. It provides a place in the implementation for outcomes of the *heofon*, *fæstenn*, and *enetre* type discussed in the literature (Fulk 2018, 121, §6.11; Ringe and Taylor 2014, 282; Campbell 1959, 155; Luick 1914--1940, 166; Hogg 1992, 112). Without an explicit rule, those outcomes would be left to diffuse analogy or to unexplained exception lists.

The consequence is therefore modest but real. The rule marks a narrow, partly lexicalized dissimilation tendency inside the larger Old English system. It does not reorganize the whole chronology, but it keeps a historically attested type of development visible in the model.

6. Remaining cautions

This section should stay short.

The evidence points to a narrow dissimilatory tendency, especially in `mn`-type clusters and a small group of lexical outcomes, rather than to a regular change operating across a broad phonological field. The rule is secure enough to model, but the available tests leave its position within the Old English sequence underdetermined.

Reader-facing pilot source note

Style-model files consulted

- `Germanic/docs/lexeme_reports/writing_skill/README.md`
- `Germanic/docs/lexeme_reports/writing_skill/book_entry_template.md`
- `Germanic/docs/lexeme_reports/model_entries/2183-shoulder-sċuldrum.model.md`

Pilot source files consulted

General sound-change sources

- `Germanic/docs/sound_changes/change_reports/full/052-velar-palatalization-hinge.md`
- `Germanic/docs/sound_changes/change_reports/full/055-056-umlaut-core.md`
- `Germanic/docs/sound_changes/change_reports/full/058-oe-nasal-dissimilation-residual-note.md`
- `Germanic/docs/sound_changes/literature_dossiers/052-velar-palatalization-hinge.dossier.md`
- `Germanic/docs/sound_changes/literature_dossiers/055-056-umlaut-core.dossier.md`
- `Germanic/docs/sound_changes/literature_dossiers/058-oe-nasal-dissimilation-residual.dossier.md`
- `Germanic/docs/sound_changes/order_tests/chronology_cards/SC052-oe-velar-palatalization.md`
- `Germanic/docs/sound_changes/order_tests/chronology_cards/SC055-oe-i-umlaut.md`
- `Germanic/docs/sound_changes/order_tests/chronology_cards/SC056-oe-ws-palatal-diphthongization.md`
- `Germanic/docs/sound_changes/order_tests/chronology_cards/SC058-oe-nasal-dissimilation.md`
- `Germanic/fsts/germanic.txt`

Reference texts checked directly

- `docs/references/campbell_old_english_grammar.txt`
- `docs/references/hogg_vol1.txt`
- `docs/references/ringe_taylor_linguistic_history_vol2.txt`
- `docs/references/luick_historische_grammatik.txt`
- `docs/references/fulk_comparative_grammar_early_germanic.vision.txt`
- `docs/references/fulk_comparative_grammar_early_germanic.pdf`

Citation verification note

For the revised pilot chapters, page citations were checked against the local text witnesses and, where useful, against the repository PDF witness as well.

Scope of the pilot

The pilot is intentionally small. It tests reader-facing section design, quotation method, code presentation, and chronology explanation before any full-volume rewrite is attempted.

Campbell, Alistair. 1959. *Old English Grammar*. Clarendon Press.

Fulk, R. D. 2018. *A Comparative Grammar of the Early Germanic Languages*. Vol. 3. Studies in Germanic Linguistics. John Benjamins.

Hogg, Richard M. 1992. *A Grammar of Old English. Volume 1: Phonology*. Blackwell.

Luick, Karl. 1914--1940. *Historische Grammatik Der Englischen Sprache*. Tauchnitz.

Ringe, Don, and Ann Taylor. 2014. *The Development of Old English*. Vol. 2. A Linguistic History of English. Oxford University Press.